Exercise 1 : SQL fundamentals

Database : Employees.db

1. SELECT Statement

Write a SQL query to receive all columns from the employees table.

```
SELECT *
from employees;
```

id	first_name	last_name	department	Salary	hire_date	city
1	John	Doe	IT	55 000	2018.06.15	New York
2	Jane	Smith	HR	48 000	2019.07.20	Chicago
3	Mike	Johnson	finance	60 000	2017.09.30	Los Angeles
4	Sarah	Brown	IT	53 000	2021.03.25	New York
5	David	White	Marketing	52 000	2016.04.10	San francisco
6	Emily	Davis	IT	62 000	2015.02.14	Chicago
7	Robert	Wilson	finance	59 000	2019.10.01	Houston
8	Jessica	Moore	HR	51 000	2018.05.22	Los Angeles
9	Daniel	Clark	Marketing	53 000	2022.06.01	Chicago
10	Laura	Hall	IT	50 000	2020.08.10	San francis

2 SELECT DISTINCT Statement.

Write SQL query to find all the unique departments in the employees table.

```
SELECT DISTINCT department
from employees;
```

department
IT
HR
finance
Marketing

3 ORDER BY Statement

Retrieve all employees' first and last names, ordered by salary in descending order

```
SELECT first_name,
       last_name,
       salary
FROM employees
ORDER BY salary DESC;
```

first_name	last_name	salary
Emily	Davis	62 000
Mike	Johnson	60 000
Robert	Wilson	59 000

4 LIMIT Statement

Write a SQL query to retrieve the top 5 highest-paid employees.

```
SELECT first_name
       last_name
       salary
FROM employees
ORDER BY salary DESC
LIMIT 5;
```

first_name	last_name	salary
Emily	Davis	62 000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55 000
Sarah	Brown	53 000

5 WHERE Statement

Write SQL query to find employees who work in the IT department.

```
SELECT first_name
       last_name
       department
FROM employees
WHERE department = 'IT';
```

first_name	last_name	department
John	Doe	IT
Sarah	Brown	IT

6. AND statement

Write a SQL query to find employees who work in the finance department AND have a salary greater than 55 000

```
SELECT first.name,
       last.name,
       department, salary
FROM employees
WHERE department = 'finance'
AND salary > 55000;
```

first.name	last.name	department	salary
Mike	Johnson	finance	60000
~~Emily~~	~~Davis~~		

7. OR statement

Write SQL query to find employees who work in HR department OR the marketing department

```
SELECT first.name,
       last.name,
       department
FROM employees
WHERE department = 'HR' OR 'marketing';
```

first.name	last.name	Department
Sane	Smith	HR
Jessica	Moore	HR
David	White	Marketing
Daniel	Clark	Marketing

8. NOT Statement

Write SQL query to find employees who do not work in the IT department

```
SELECT first.name,
       last.name,
       department
FROM employees
WHERE NOT department = 'IT';
```

first.name	last.name	department
Sane	Smith	HR
Mike	Johnson	finance
David	White	Marketing
Robert	Wilson	finance
Jessica	Moore	HR

9. IN Statement

Write a SQL query to find employees who are in the HR, IT or finance departments.

```
SELECT first_name,
       last_name,
       department
FROM employees
WHERE department IN 'HR','IT','finace'
```

first_name	last_name	department
John	Doe	IT
Jane	Smith	HR
Mike	Johnson	finance
Sarah	Brown	IT
Emily	Davis	IT
Robert	Wilson	finance
Jessica	Moore	HR
Laura	Hall	IT

10. Combining Conditions

Write a SQL query to find employees who are in the IT department, have a salary greater than 50 000 and are located in NY.

```
SELECT first_name,
       last_name,
       department
FROM employees
WHERE department = 'IT'
AND salary > 5 000
AND city NEW YORK
```

first_name	last_name	dept.	salary	city
John	Doe	IT	55000	NY
Sarah	Brown	IT	53000	NY
Emily				

11. Combining WHERE, AND and ORDER BY

Write SQL query to retrieve the first and last names of employees who work in the finance or Marketing department, earn more than 52 000 and order the results by salary in descending order.

```
SELECT first_name,
       last_name
       department,
       salary
FROM employees
WHERE (department = 'finance' OR 'Marketing'
AND salary > 52 000
ORDER BY salary DESC;
```

first_name	last_name	department	salary
Mike	Johnson	finance	60 000
Robert	Wilson	finance	54 000
Daniel	Clark	Marketing	53 000
David	White	Marketing	52 000

12. Combining SELECT DISTINCT, WHERE and IN
Write SQL query to find all unique cities where employees work, excluding those in IT And HR departments.

```
SELECT DISTINCT city
FROM employees
WHERE department NOT IN 'IT', 'HR');
```

department
finance
Marketing

13. Combining WHERE NOT AND ORDER BY
Write SQL query to retrieve employees who are NOT in the finance department, have salary greater than 50 000 and order the results by hire date in ascending order.

```
SELECT first_name,
       last_name,
       department,
       salary,
       hire_date
FROM employees
WHERE department = 'finance'
AND salary > 50 000
ORDER BY hire_date DESC;
```

first_name	last_name	department	salary	hire_date
David	White	Marketing	52 000	2016-04-10
Jessica	Moore	HR	51 000	2018-03-22
Daniel	Clark	Marketing	55 000	2022-06-01

14. Combining WHERE, OR, IN & LIMIT

Write a SQL query to find the first three employees who work either Chicago or Los Angeles and belong to the IT or Marketing department

```
SELECT first_name, last_name, department, salary, city
FROM employees
WHERE (department) = 'IT' OR finance')
AND city' = San fransisco'
AND salary > 55 000
ORDER BY salary DESC
LIMIT 5;
```

```
SELECT first_name,
       last_name,
       department,
       city
FROM employees
WHERE city IN ('Chicago', 'Los Angeles')
AND department IN ('IT', 'Marketing')
LIMIT 3;
```

first_name	Last_name	department	city
David	White	Marketing	Chicago
Daniel	Clark	Marketing	Chicago
Emily	Davis	IT	Chicago